Rates of Reaction Factors

Exam code: 0620 & 0971

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Written by: Alexandra Brennan
Reviewed by: Stewart Hird
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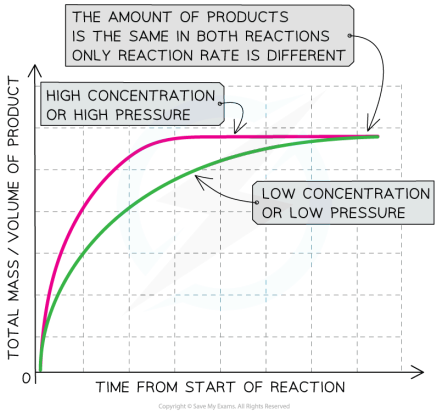
Rates of reaction factors

- Factors that can affect the rate of a reaction are:
  - The **concentration** of the reactants in solution or the **pressure** of reacting gases
  - **Surface area** of solid reactants
  - The **temperature** of the reaction
  - The presence of a **catalyst**
- Changes in these factors directly influence the rate of a reaction
- It is of **economic interest** to have a higher rate of reaction as this implies a higher rate of production and hence a more efficient and sustainable process

The effect of increased concentration or pressure

Graph showing the effect of concentration on rate of reaction



Increasing the concentration of a solution or gas pressure increases the rate of reaction

Explanation:

- Compared to a reaction with a reactant at a low concentration (or pressure), the line graph for the same reaction at a higher concentration (or pressure):
  - Has a steeper gradient at the start
  - Becomes horizontal sooner

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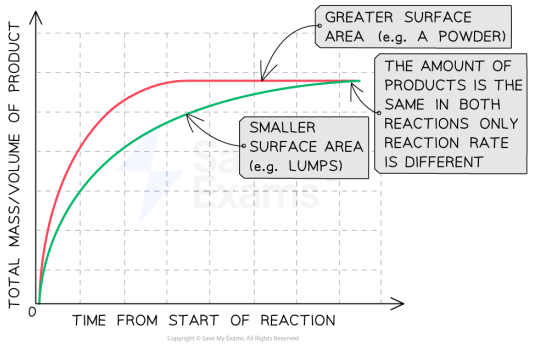
The effect of increasing surface area

Graph showing the effect of surface area on rate of reaction

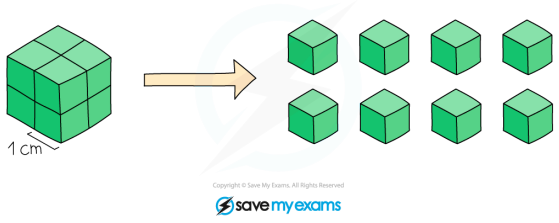


Increasing the surface area increases the rate of reaction

Explanation:

- Compared to a reaction with lumps of reactant, the line graph for the same reaction with powdered reactant:
  - Has a steeper gradient at the start
  - Becomes horizontal sooner
  - Forms the same amount of product
- This shows that **increasing the surface area increases the rate of reaction**
  - Increasing surface area can sometimes be described as decreasing solid particle size

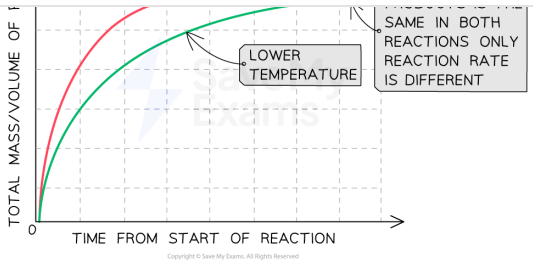
Surface area and particle size



Surface area increases as particle size decreases. A 2 cm cube has a surface area of 24 cm^2 and the same cube cut up into 8 cubes has a surface area of 48 cm^2

The effect of increasing temperature

Graph showing the effect of temperature on rate of reaction



Increasing the temperature increases the rate of reaction

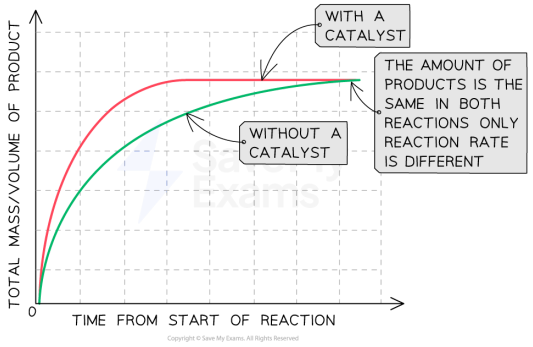
Explanation:

- Compared to a reaction at a low temperature, the line graph for the same reaction at a higher temperature:
  - Has a steeper gradient at the start
  - Becomes horizontal sooner
  - Forms the same amount of product
- This shows that **increasing the temperature increases the rate of reaction**

The effect of using a catalyst

- Catalysts are substances which **speed up** the **rate** of a reaction without themselves being **altered** or **consumed** in the reaction
- The mass of a catalyst at the beginning and end of a reaction is the **same** and they do not form part of the equation

Graph showing the effect of using a catalyst on rate of reaction



Graph showing the effect of using a catalyst on the rate of reaction

Explanation:

- Compared to a reaction without a catalyst, the graph line for the same reaction but with a catalyst has a steeper gradient at the start and becomes horizontal sooner
- **This shows that with a catalyst, the rate of reaction will increase**

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